

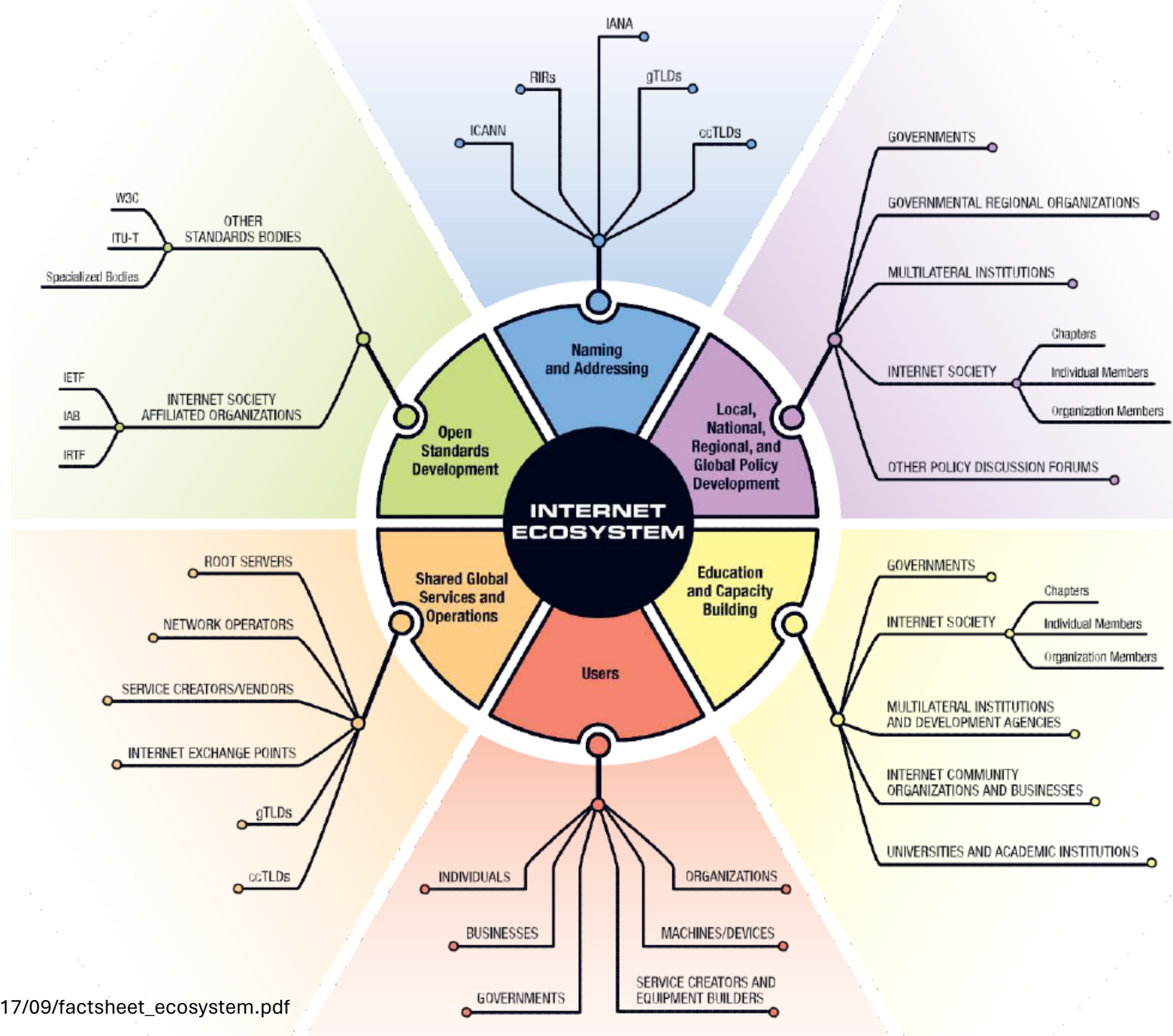
Ekosistem Internet

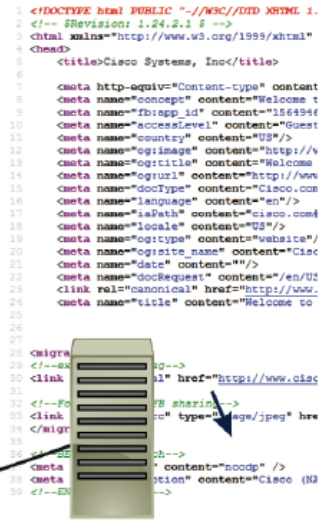
Dr. Ferry Astika Saputra

Bagaimana Internet bisa berkembang dengan pesat di seluruh dunia ?

- Tiga Kunci sukses Internet:
 - kepemilikan global Bersama
 - pengembangan standar terbuka
 - proses pengembangan teknologi dan kebijakan yang dapat diakses secara bebas.
- Kesuksesan Internet yang belum pernah terjadi sebelumnya terus berkembang karena model Internet yang terbuka, transparan, dan kolaboratif → mengandalkan proses dan produk yang bersifat lokal, bottom-up, dan dapat diakses oleh pengguna di seluruh dunia.

Ekosistem Internet





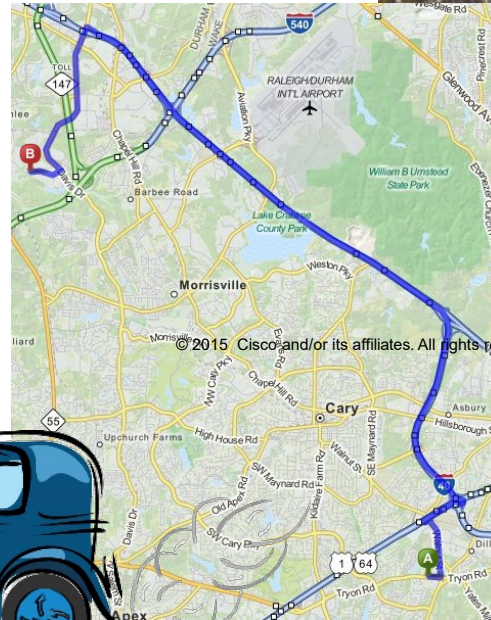
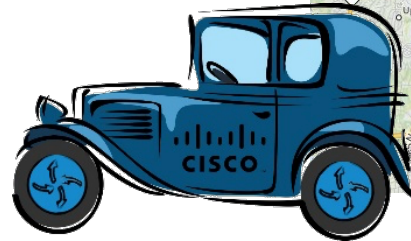
Internet dapat dipandang dari 2 sisi

- Teknis
 - *Routing system* (system routing)
 - *Naming system* (sistem penamaan)
- Arsitektur
 - *Standards* (standarisasi)
 - *Service Providers* (penyedia layanan)
 - *Internet Registries* (Internet register)
 - *Clearing Houses* (Rumah kliring)

IP Addressing & Routing System

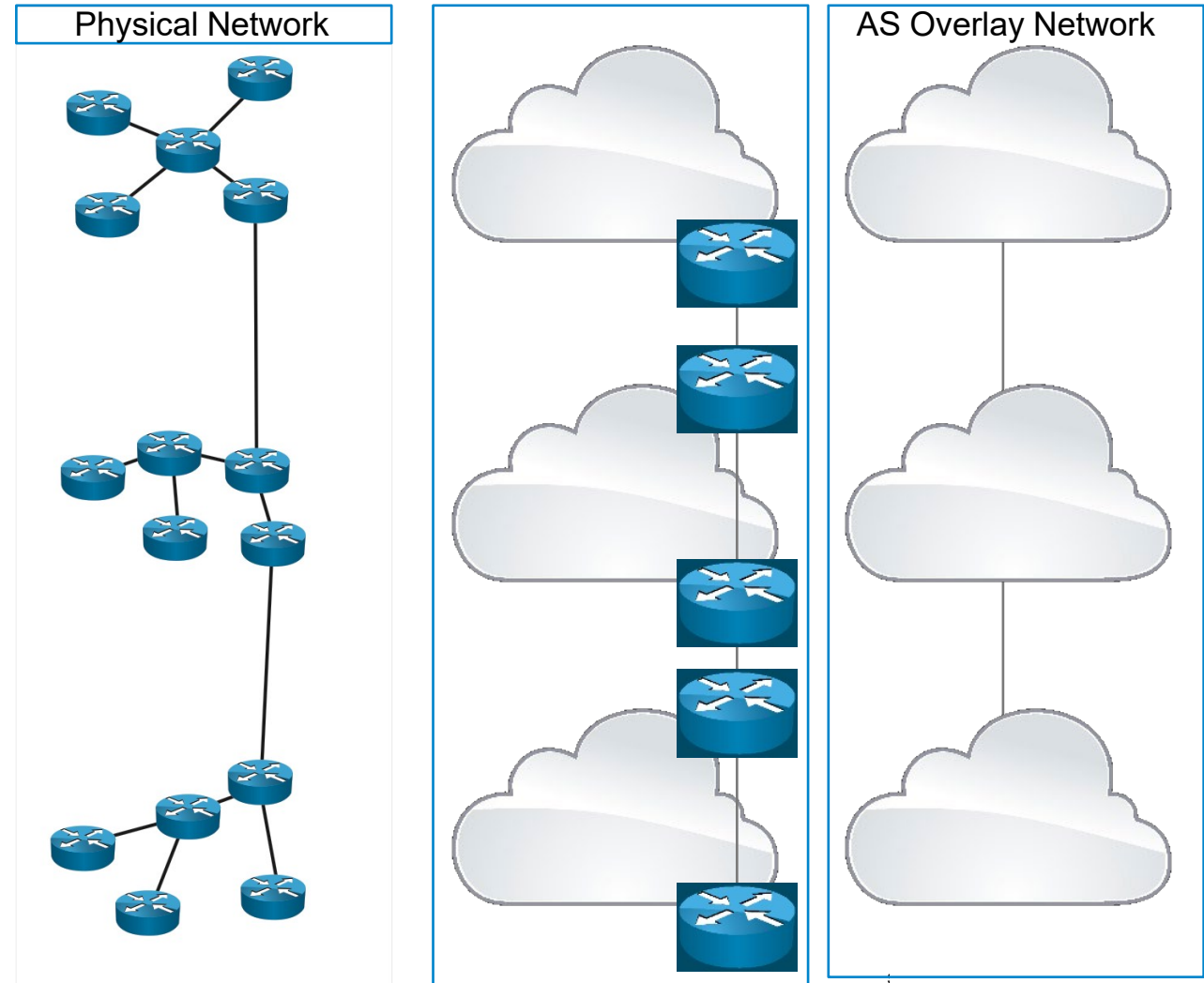
Routing

- Now we have an address that tells us where to find the web site we're after...
- And we have a way to bundle the data that needs to be transferred...
- But how do we get there?



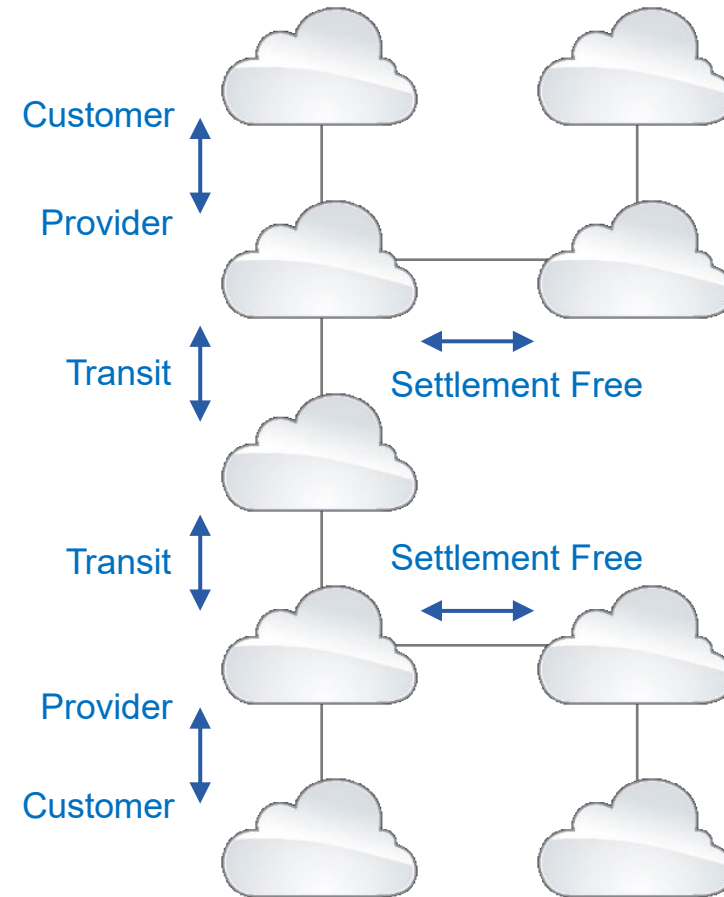
Routing

- It's easy to say “the Internet is a network of networks,” but what does that really mean?
- Another way to view the Internet is as a virtual topology
- Policy is the key to understanding the AS level (overlay network)
 - BGP is *distributed policy*
 - Different types of providers follow different sets of policies to maximize revenue and minimize cost



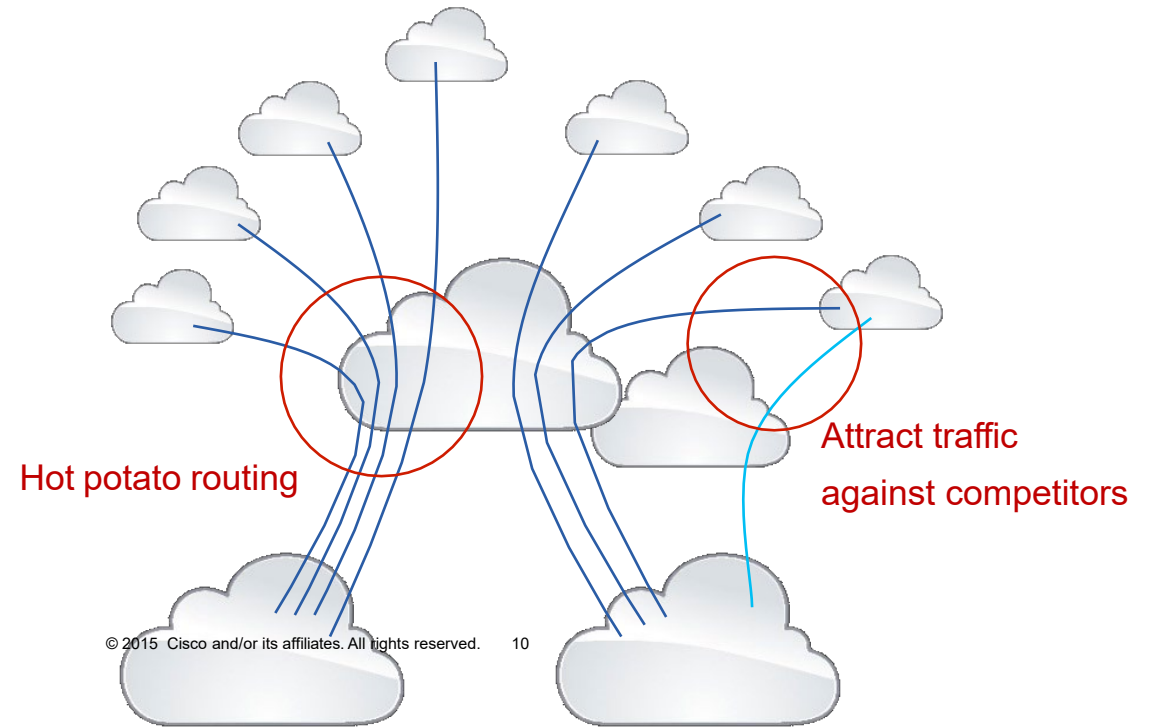
Peering Connections

- Understanding these policies requires understanding peering models
- Provider/Customer
- Transit
- Settlement Free



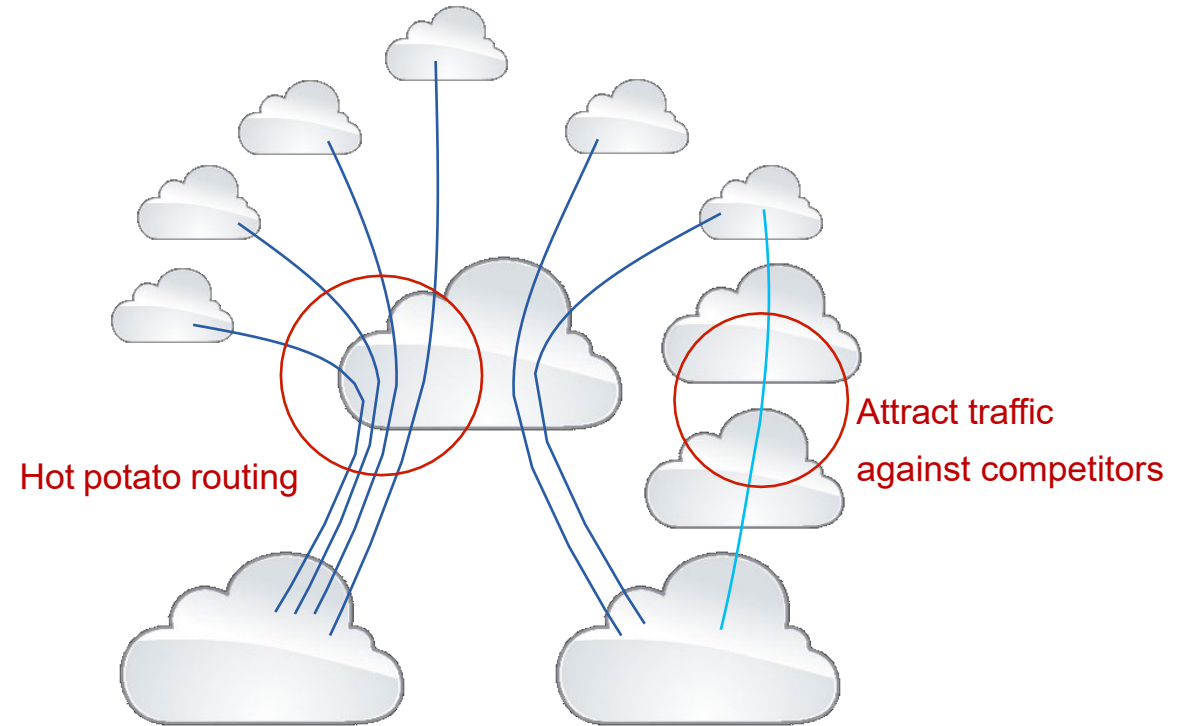
Edge Provider Routing Policy

- Be the preferred path for connected customers
 - Even if they're connected to someone else
- Carry traffic across the shortest possible path
 - Or for the least amount of time possible
- “Hot potato routing”



Transit Provider Routing Policy

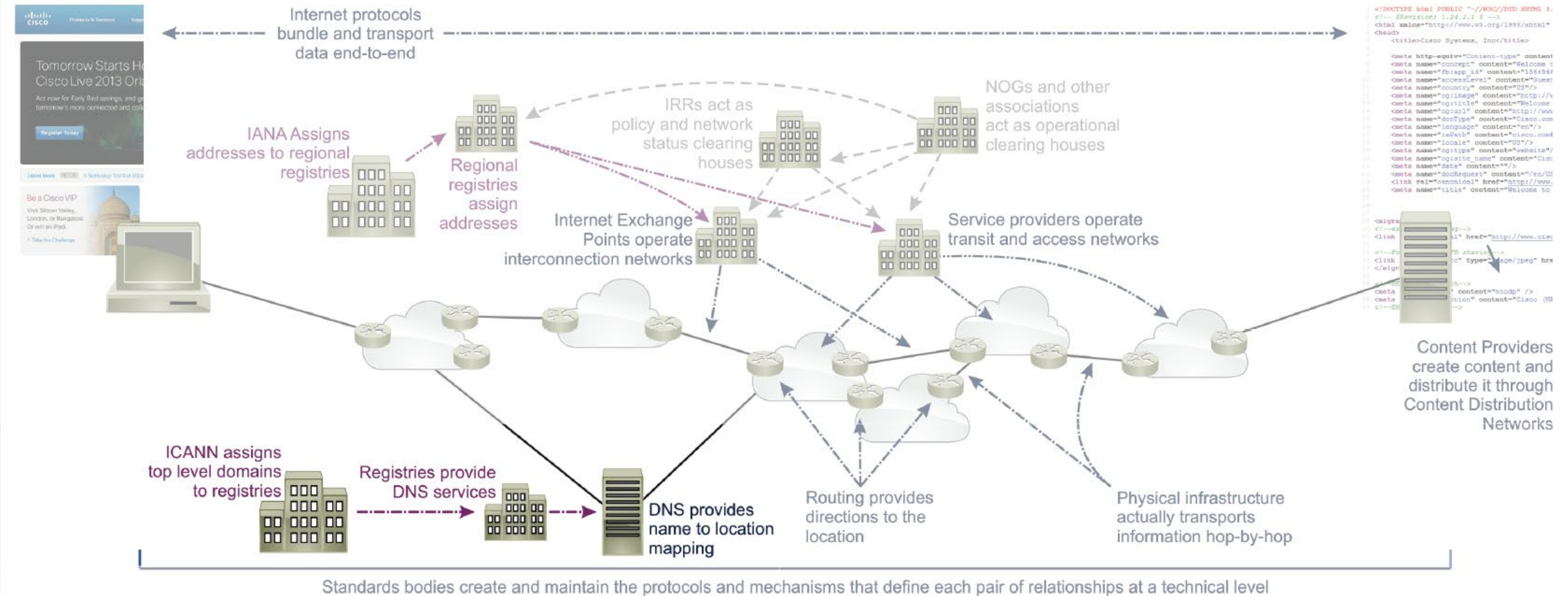
- Attract as much traffic as possible
 - Increasing peering to reduce the AS Path length to reach any particular destination
- Carry as little traffic as possible
 - Really carrying packets for the shortest distance you can
 - “Hot potato routing”



Content Provider Routing Policy

- Push content as close to the actual “eyeballs” as possible
 - Peer as widely as possible
 - Use content distribution mechanisms to push content to every corner/edge of the network
- Carry traffic on internal links where possible
 - To exert the greatest level of control over the user’s experience on your services
 - “Cold potato routing”

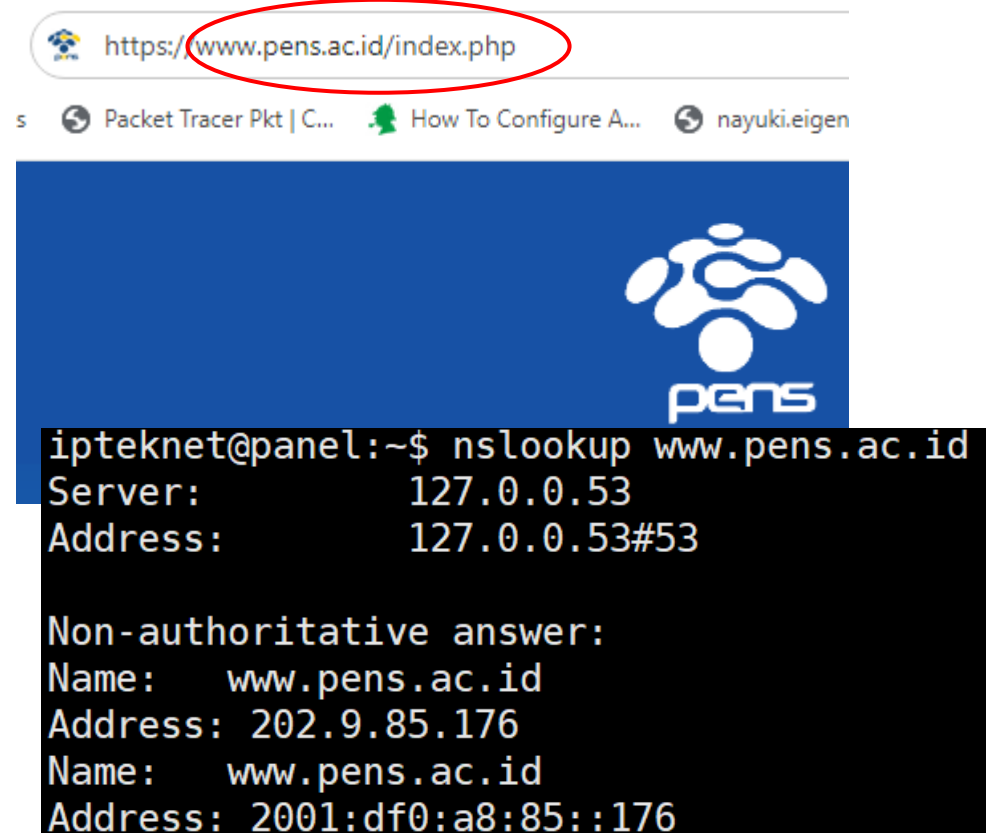
Naming System



Naming System

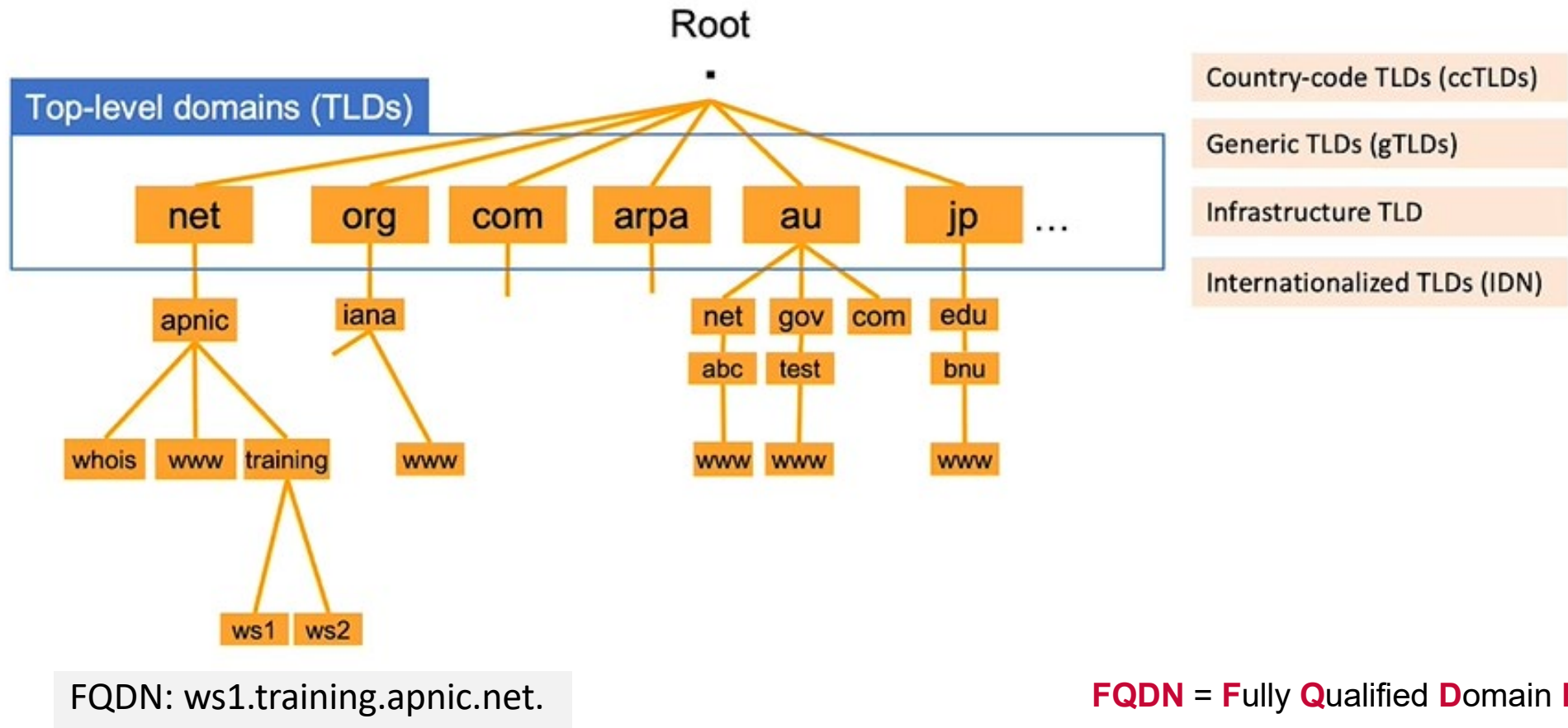
Domain Name System (DNS)

- Dimulai dengan pengguna yang mengetikkan nama situs web ke dalam browser di sebuah host
- Bagaimana host menemukan situs web berdasarkan nama?
- Ada tiga hal yang perlu dilakukan host:
 - menemukan alamat server tempat informasi berada
 - Bagaimana menuju ke alamat tersebut
 - File spesifik di server itu sendiri
- DNS menyediakan translasi dari IP Address ke NAMA_SERVER/HOST



www.pens.ac.id/index.php = www (nama server) + pens.ac.id (nama domain) + nama file (landing page/home page)

DNS Hierarchy Tree



DNS Components

Namespace

Domains and zones

Nameserver

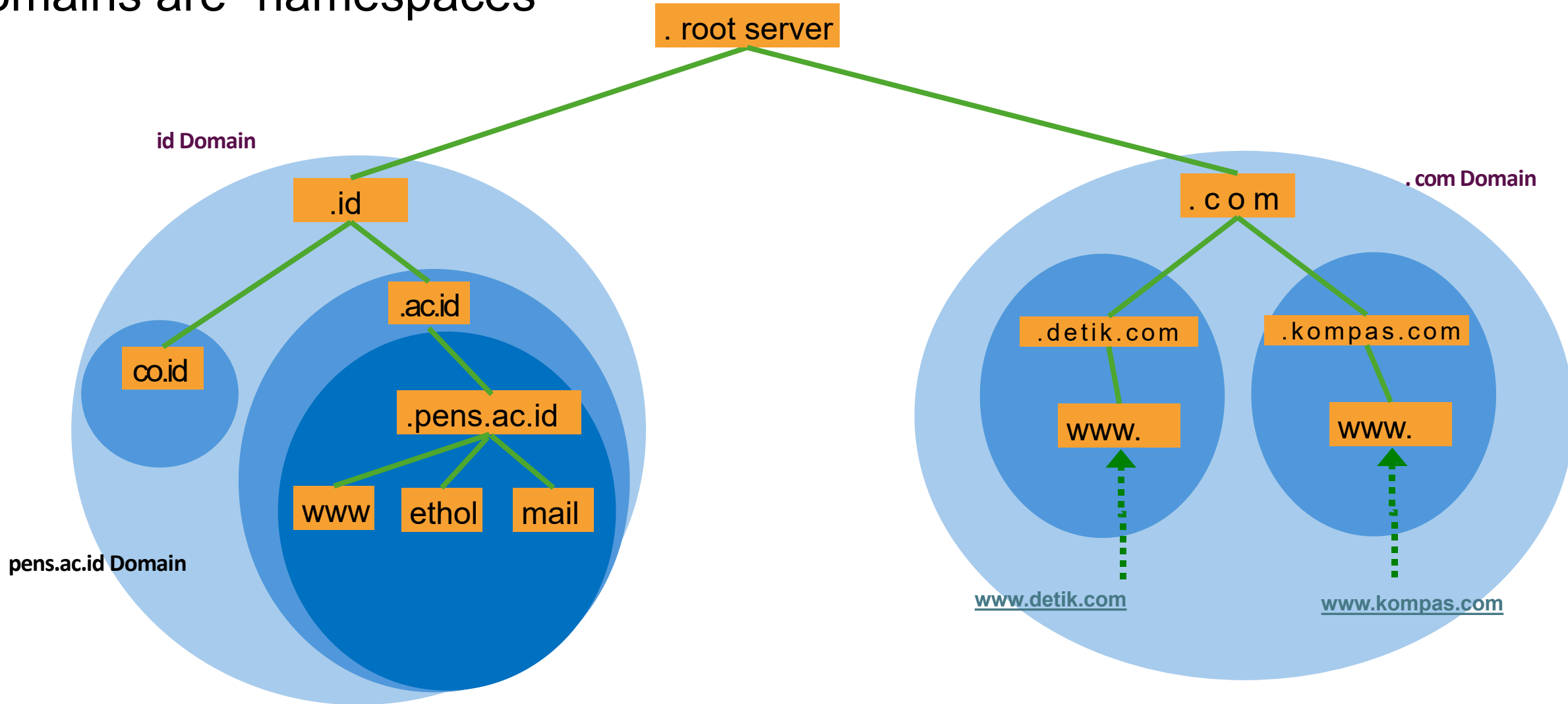
Makes the namespace available

Resolvers or clients

Query the name server for records in the namespace

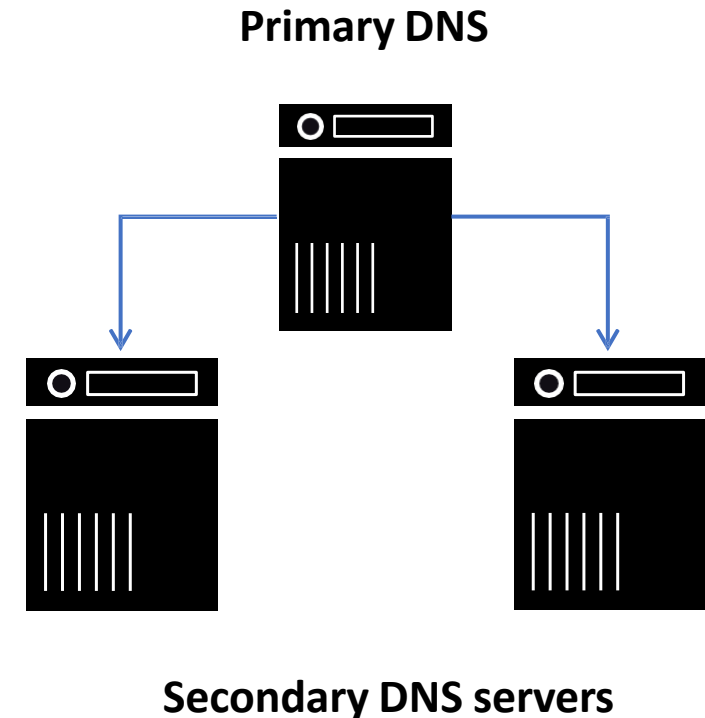
Domains

Domains are “namespaces”



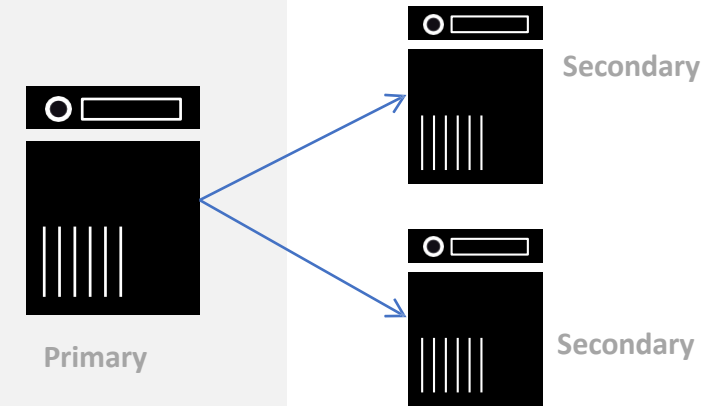
Name Servers

- Name servers answer DNS questions
- Several types of name servers
 - Authoritative servers
 - Primary
 - Secondary
 - Recursive servers
 - also caching forwarders
- Mixture of functions



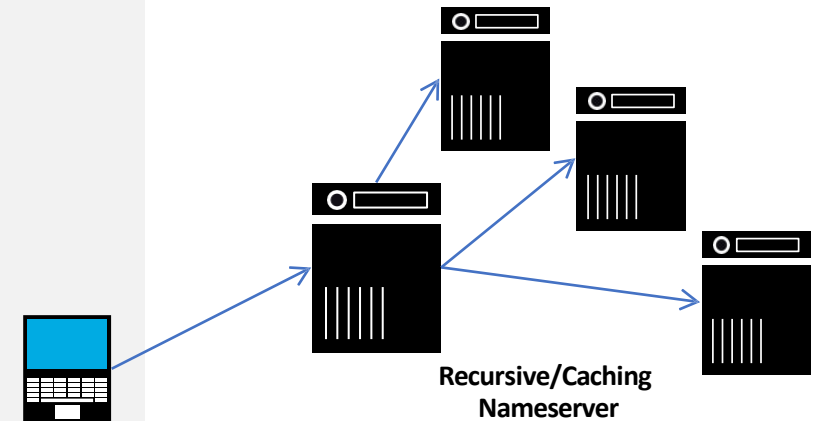
Authoritative Nameserver

- A nameserver that is authorised to provide an answer for a particular domain
 - Can be more than one auth nameserver
- Two types based on management method:
 - Primary (Master) and Secondary (Slave)
- Only one primary nameserver
 - All changes to the zone are done in the primary
- Secondary nameserver/s will retrieve the zonefile from the primary server
 - Secondary polls the primary periodically
- Primary server can “notify” the secondary servers



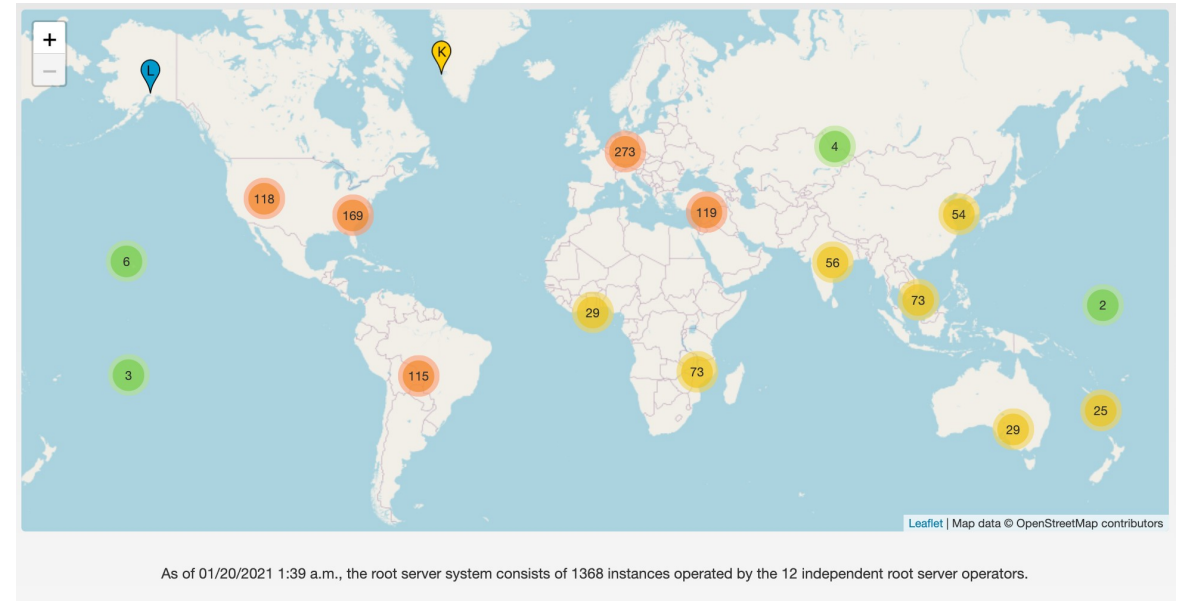
Recursive Nameserver

- The job of the recursive nameserver is to locate the authoritative nameserver and get back the answer
- This process is iterative – starts at the root
- Recursive servers are also usually caching servers
- Prefer a nearby cache
 - Minimizes latency issues
 - Also reduces traffic on your external links



Root Servers

- The top of the DNS hierarchy
- There are 13 root name servers operated around the world
`[a-m] .root-servers.net`
- There are more than 13 physical root name servers
 - Each rootserver has an instance deployed via anycast



Root Server Deployment at APNIC

- Started in 2002, APNIC is committed to establish new root server sites in the AP region
- The aim is to strengthen DNS by deploying additional resources to handle growing Internet traffic.

Timeline of root server deployment

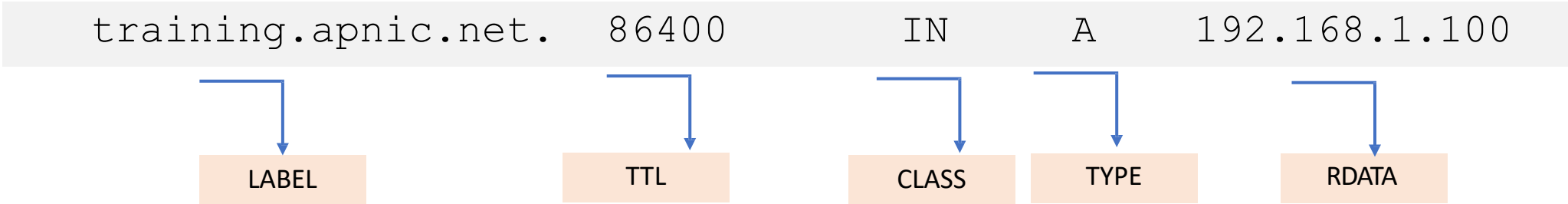
2020	December M-Root nameserver installed in Brisbane.
2019	January K-Root nameserver installed in Thimphu. December K-Root nameserver installed in Yangon.
2018	July F-Root nameserver installed in Port Moresby. December K-Root nameserver installed in Taipei.
2017	January J-Root nameserver installed in Kathmandu.

Ref: <https://www.apnic.net/community/support/root-servers/>

Resource Records

Entries in the DNS zone file

Resource Record	Function
Label	Name substitution for FQDN
TTL	Timing parameter, an expiration limit
Class	IN for Internet, CH for Chaos
Type	RR Type (A, AAAA, MX, PTR) for different purposes
RDATA	Anything after the Type identifier; Additional data



Common Resource Record Types

RR Type	Name	Functions
A	Address record	Maps the domain name to IP address www.example.com. IN A 192.168.1.1
AAAA	IPv6 address record	Maps the domain name to an IPv6 address www.example.com. IN AAAA 2001:db8::1
NS	Name server record	Used for delegating zone to a nameserver example.com. IN NS ns1.example.com.
PTR	Pointer record	Maps an IP address to a domain name 1.1.168.192.in-addr.arpa. IN PTR www.example.com.
CNAME	Canonical name	Maps an alias to a hostname web IN CNAME www.example.com.
MX	Mail Exchanger	Defines where to deliver mail for user @ domain example.com. IN MX 10 mail01.example.com. IN MX 20 mail02.example.com.

Example: RRs in a Zone File

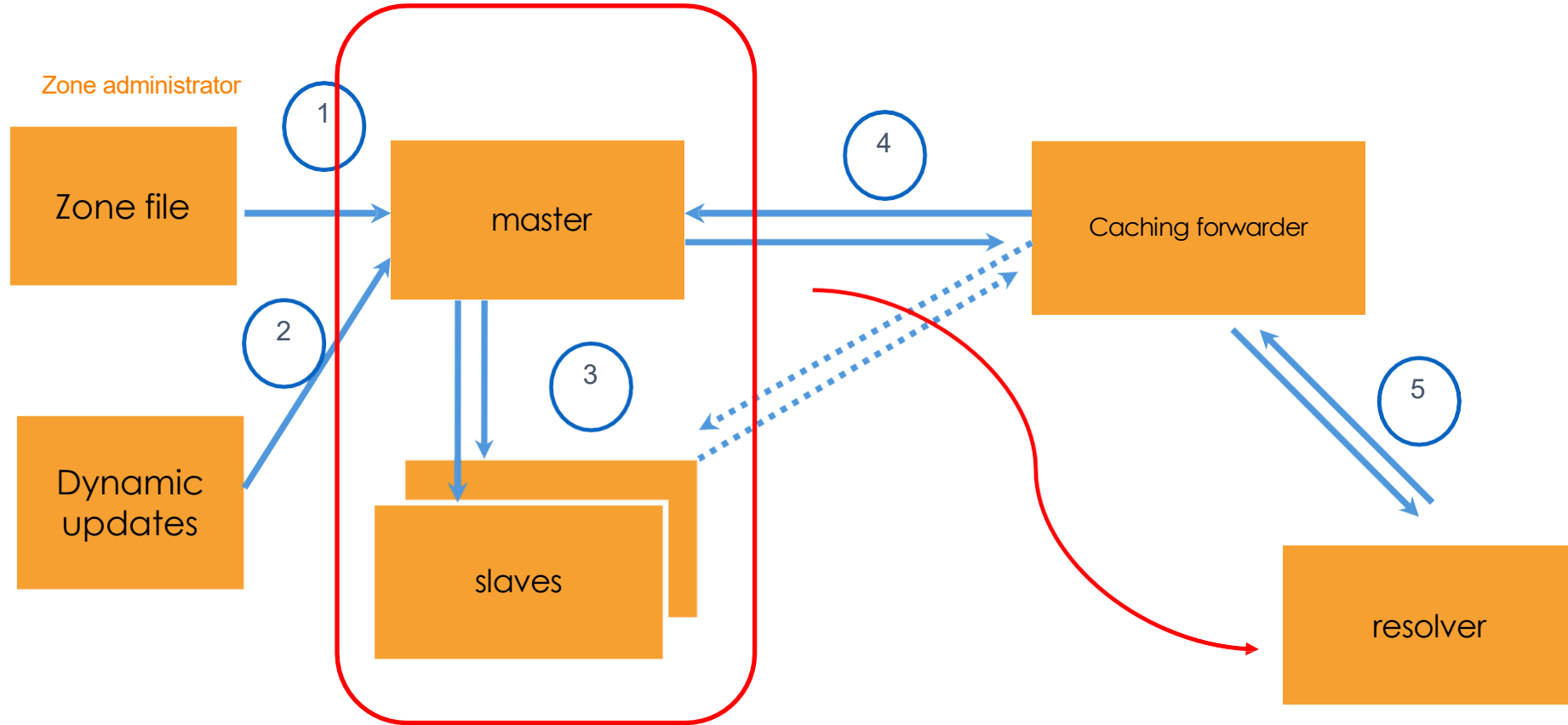


```
apnic.net.      7200      IN      SOA      ns.apnic.net. admin.apnic.net. (
                2020072001    ; Serial
                12h          ; Refresh 12 hours
                4h           ; Retry 4 hours
                4d           ; Expire 4 days
                2h )         ; Negative cache 2 hours
```

```
apnic.net.      7200      IN      NS              ns.apnic.net.
apnic.net.      7200      IN      NS              ns.ripe.net.
```

```
www.apnic.net. 3600      IN      A              192.168.0.2
www.apnic.net 3600      IN      AAAA         2001:DB8::2
```

DNS Data Flow



Delegating a Zone

Delegation is done by adding NS records.

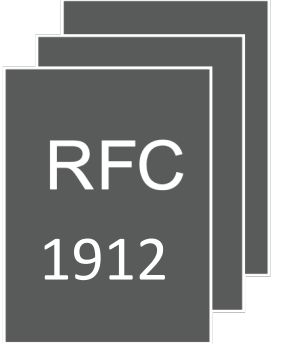
In this example, **apnic.net** zone is delegating the subdomain **academy.apnic.net** to these 2 nameservers.

```
;From apnic.net zone, add these records:
```

```
academy.apnic.net.    NS      ns1.academy.apnic.net.  
academy.apnic.net.    NS      ns2.academy.apnic.net.
```

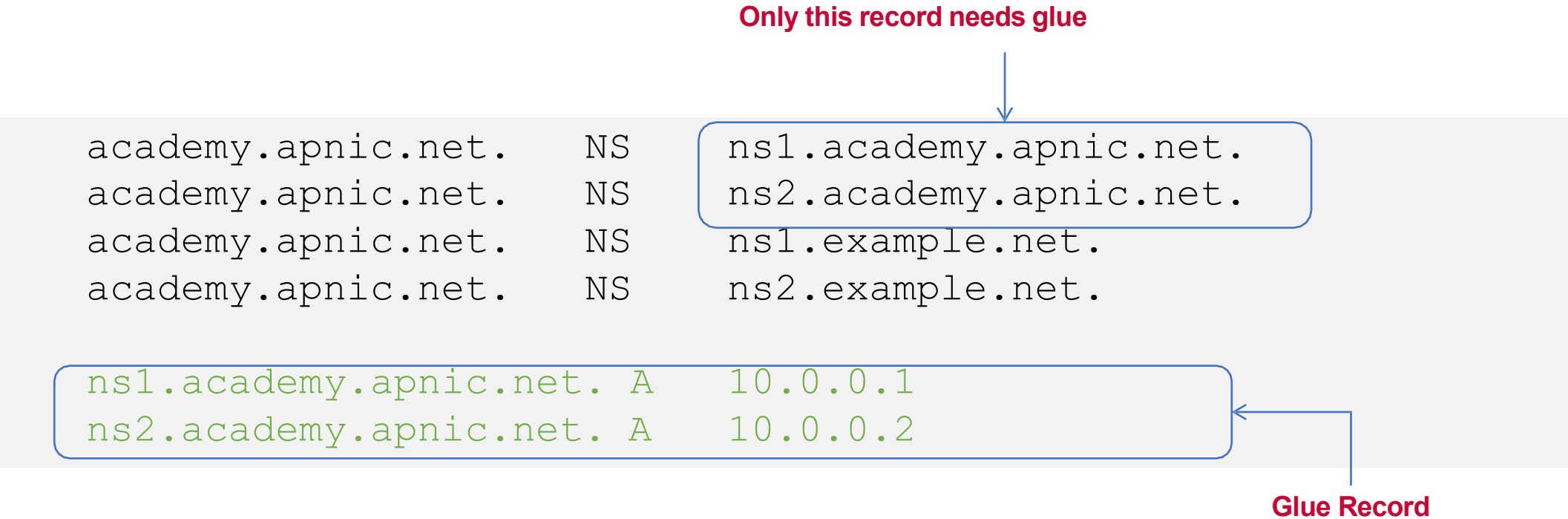
A client must then go to ns1.academy.apnic.net (or ns2) to query for any of its subdomain.

Now how can we reach ns1 and ns2? We must add a **Glue Record**.



Glue Record

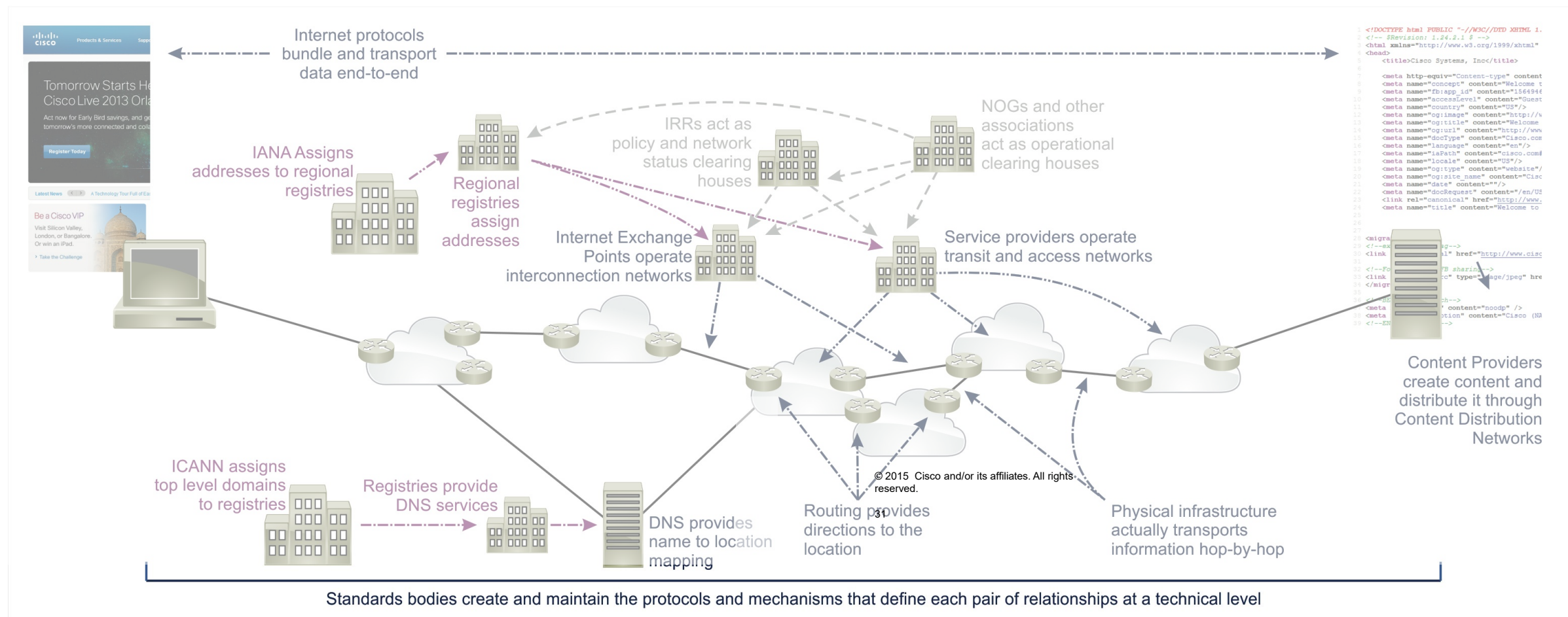
A **glue record** is a non-authoritative data. It is an A record that maps the address of the sub-domain's nameserver.



STANDARDS BODIES

(Lembaga Standarisasi)

Standards Organizations



Standards Organizations



I E T F[®]

All layer 3 transport and routing protocols, including IP, TCP, UDP, HTTP, NS, routing protocols, telnet, rsync, IPsec, and network management protocols



IEEE

All layer 1 and layer 2 transport and control plane protocols, including Ethernet, spanning tree, and wireless networking



Markup languages (languages which describe how to display or render content), including HTML and XML



Any international standard, including numbering, encryption schemes, and routing protocols (such as IS-IS)

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IETF History

- Formed in 1986
 - Evolved out of US ARPANET-related government activities
 - Internet Configuration Control Board (ICCB) (1979) and Internet Activities Board (1983)
- People not companies are participants!
- Has developed or maintains most Internet-related standards.
- Does not exist (in a legal sense), no members, no voting
 - The IETF is “an organized activity of the Internet Society”

Internet Engineering Task Force (IETF)

- Mission: to make the Internet work better.
- Role and Scope
 - *‘above the wire and below the application’*
 - IP, TCP, email, routing, IPsec, HTTP, FTP, ssh, LDAP,
 - SIP, mobile IP, ppp, RADIUS, Kerberos, secure email,
 - streaming video & audio, ...

IETF Meetings

IETF 95

April 3-8, 2016
Buenos Aires, Argentina

Recent Meetings

- **IETF 92**
 - March 22-27, 2015
 - Dallas, TX, USA
- **IETF 93**
 - July 19-24, 2015
 - Prague, Czech Republic
- **IETF 94**
 - November 1-6, 2015
 -  Yokohama, Japan

Upcoming Meetings

- **IETF 96**
 - July 17-22, 2016
 - Berlin, Germany
- **IETF 97**
 - November 13-18, 2016
 - Seoul, South Korea
- **IETF 98**
 - March 26-31, 2017
 - Chicago, IL, USA

IETF Organization: Areas

General Area (gen)

- ...activities focused on supporting, updating and maintaining the IETF standards development process.

Security (sec)

- ...focused on security protocols...services: integrity, authentication, non-repudiation, confidentiality, and access control...key management is also vital.

Applications and Real Time (art)

- Protocols for delay-sensitive communications, and building blocks to be used across a wide variety of applications.

Operations & Management (ops)

- Network Management, AAA, and various operational issues facing the Internet such as DNS, IPv6, operational security and Routing operations.

Transport Services (tsv)

- ...works on mechanisms related to end-to-end data transport...

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Routing (rtg)

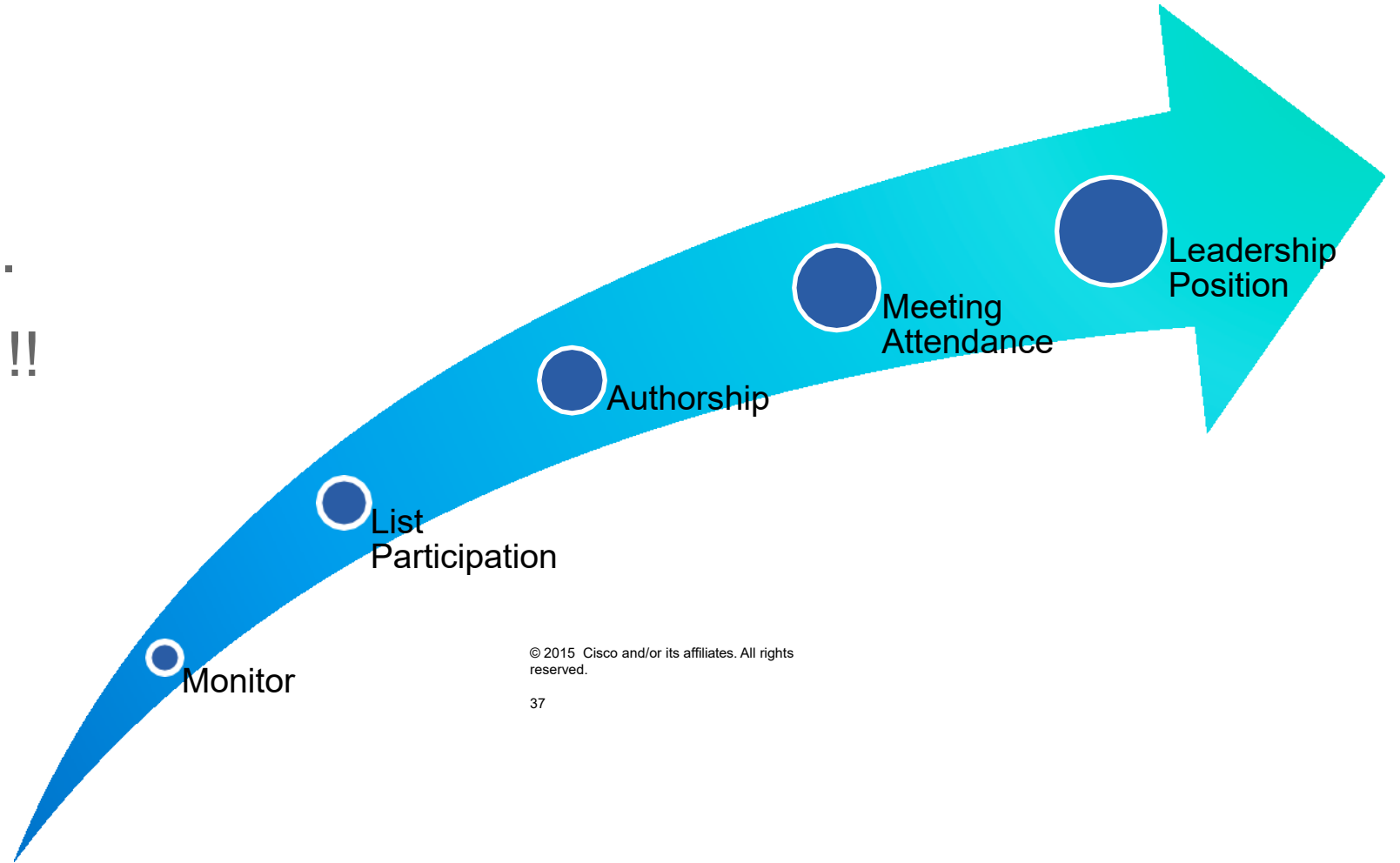
- ...responsible for ensuring continuous operation of the Internet routing system...

Internet (int)

- ...IP layer (both IPv4 and IPv6), DNS, mobility, VPNs and pseudowires..., and various link layer technologies.

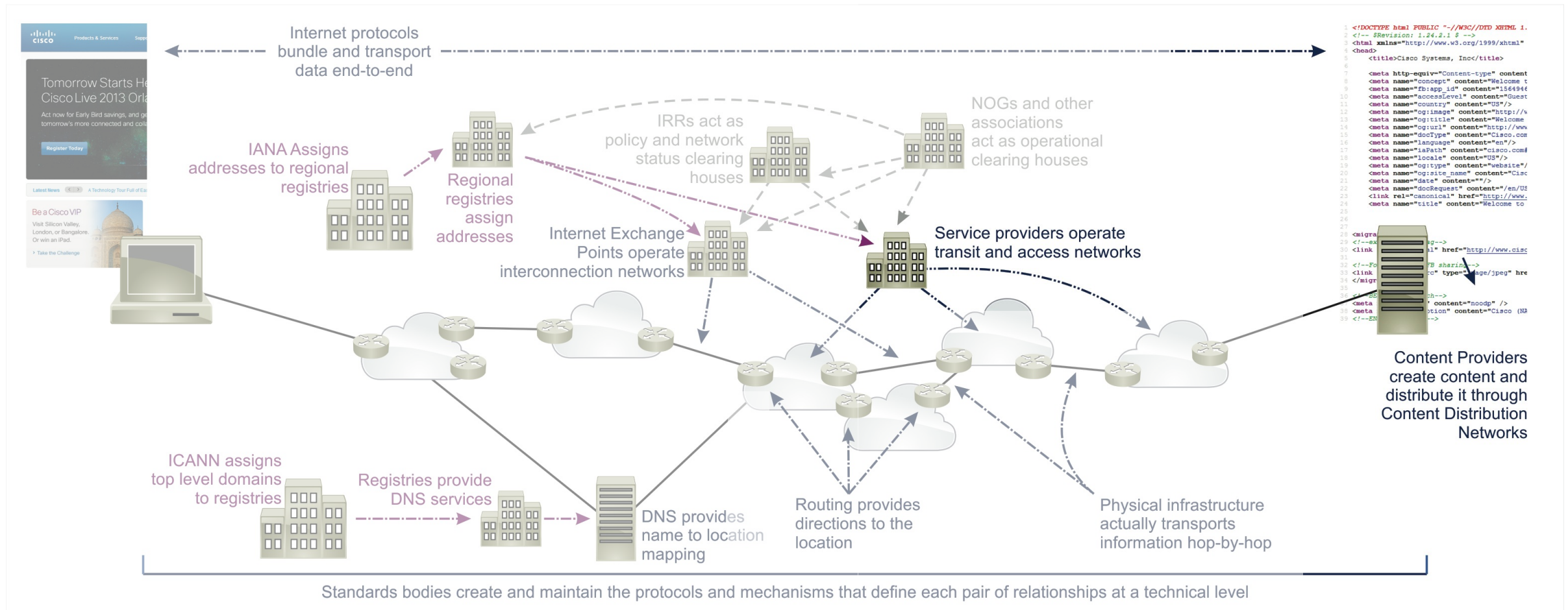
How to Participate in the IETF?

- Join a mailing list..
- ..start contributing!!



SERVICE PROVIDERS
(PENYEDIA LAYANAN)

Service Providers



Content Provider Overview

- Content providers fall into two categories
 - Media creation and distribution
 - Connecting buyers and sellers (ecommerce)
- Sell services/entertainment or connect buyers to sellers.
 - User attention (advertising).



Ecommerce, entertainment, information



Search, services, software (ecommerce)



Search, services, entertainment



Entertainment



Services, information, ecommerce



Entertainment



Entertainment



Ecommerce

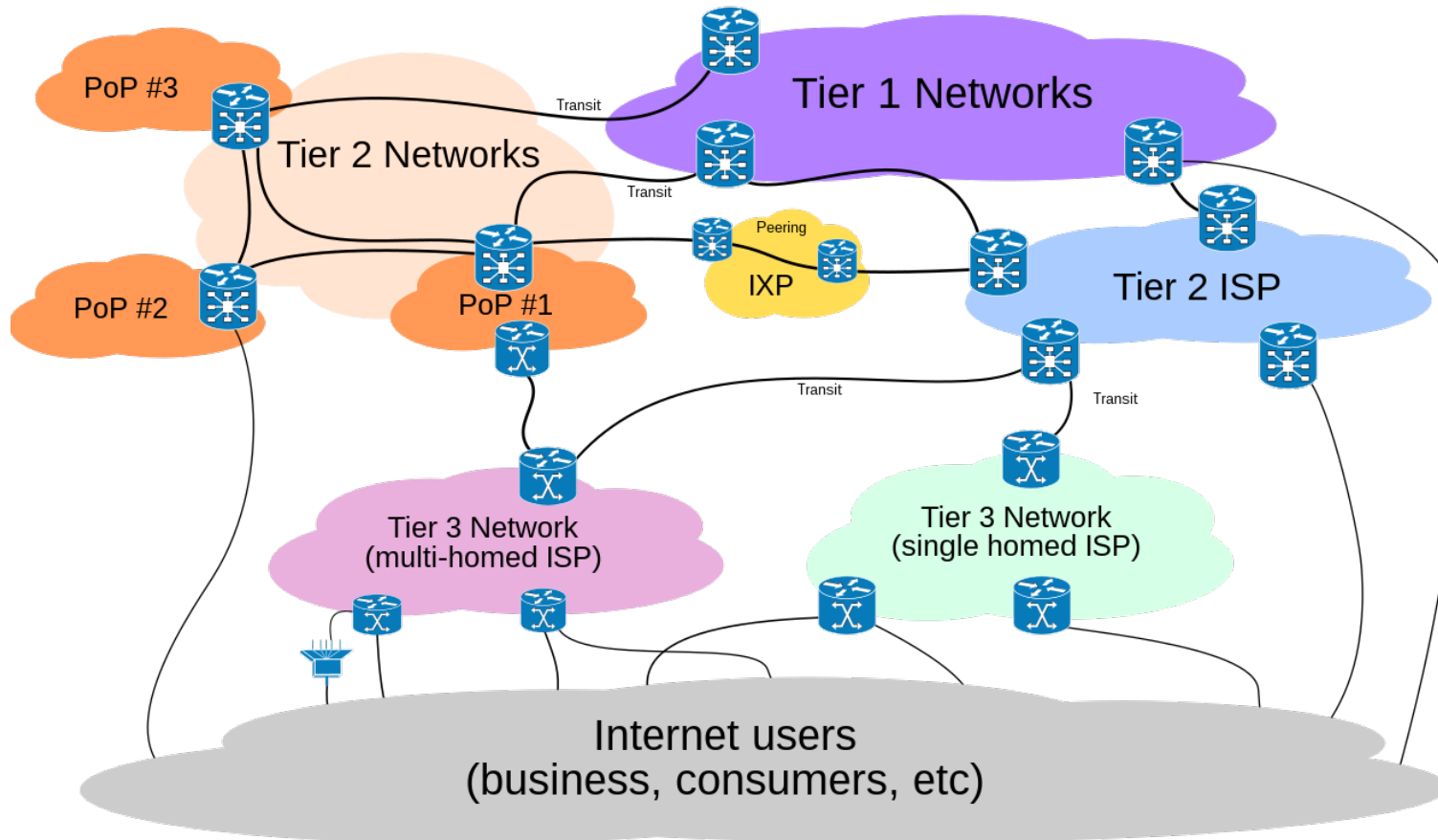
Access Provider Overview

- Provide individual users, businesses, and organizations with Internet connections
- Are often involved in the creation and distribution of content, as well

	Wireless services, business Internet access
	High speed Internet access
	Satellite based Internet access
	Internet Mobile Services
	Fiber to the Home Services
	High Speed Internet Access

Transit Provider Overview

- Provide interconnection between content and access service providers

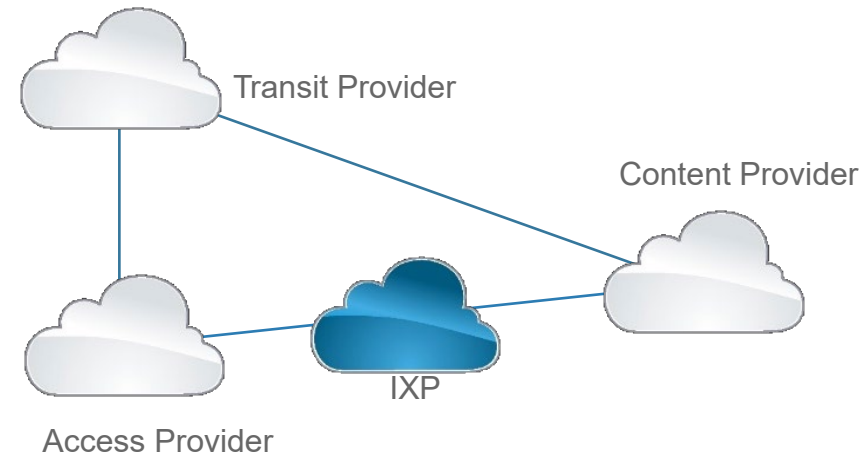


https://en.wikipedia.org/wiki/Tier_1_network

https://en.wikipedia.org/wiki/Tier_2_network

Internet Exchange Point Overview

- Internet Exchange Points (IXPs) provide local interconnections for access and content providers
 - IXPs can either be commercial or non-profit
- Used by providers within a region
 - Avoids the use of out of region transit providers
 - Reduces latency within the region
 - Replaces traffic based settlement costs with fixed membership fees

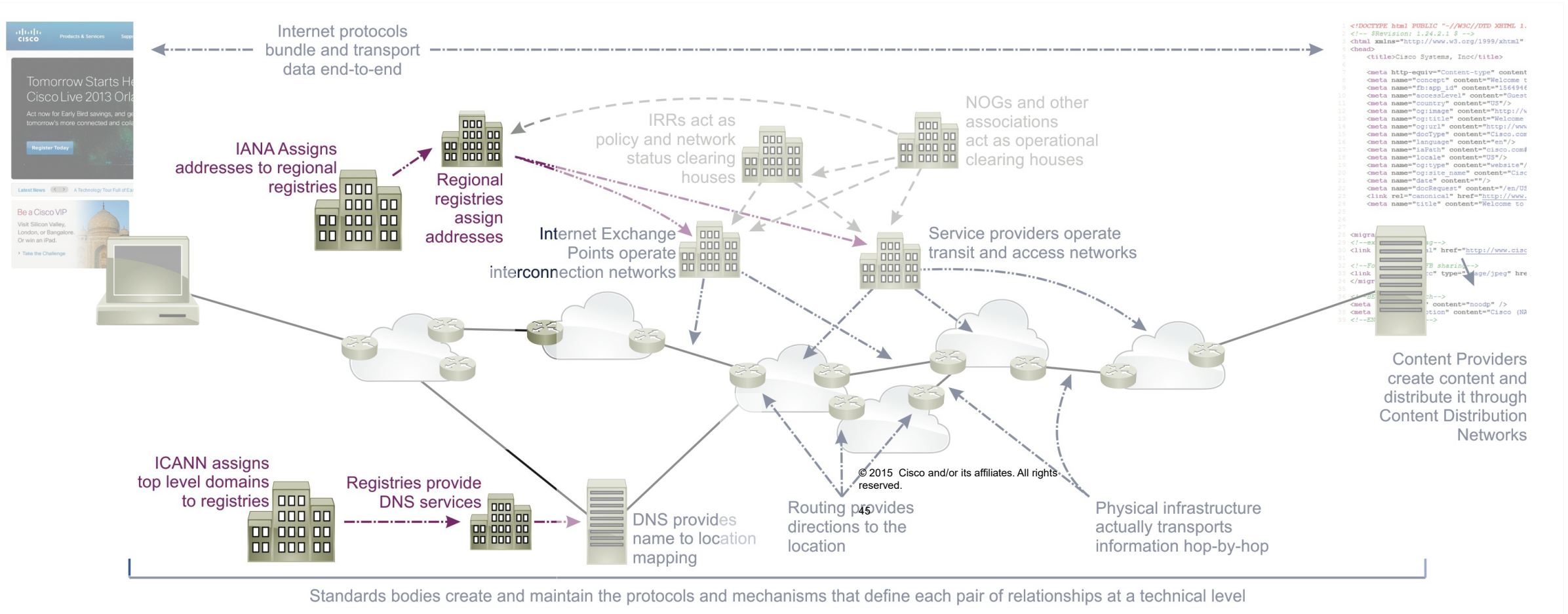


openixp



REGISTRIES

Registries



Naming Authorities

- ICANN and IANA coordinate the assignment of the numbers and names that make the Internet work
 - They only set the rules under which other organizations may obtain these resources
- IANA manages much more than the IP address space



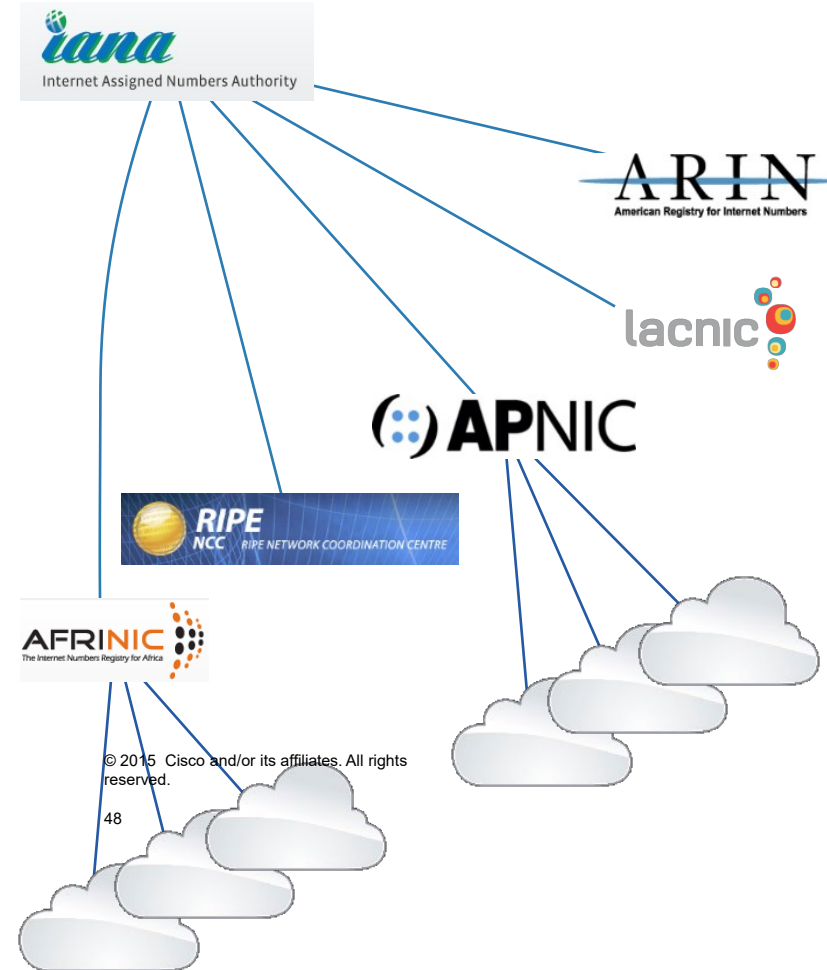
Regional Registry Overview

- Regional Internet Registries (RIRs) manage IP address block assignments for a region
- RIRs also...
 - Participate in research and standardization efforts
 - Provide services for the “common good” of their members, such as *whois*
 - Hold regular meetings for the education of their members and information exchange
- Produce best common practice documents and recommendations



Regional Registry Overview

- IANA provides blocks of IP addresses to each RIR
 - Delegation is primarily based on need
- Each regional registry then provides address blocks to their members
 - The members can then advertise this space in the global routing table, use them in DNS entries, or even use them just for their internal networks



Top Level Registries

- Domain Name Registries manage Top Level Domains (TLDs)
 - ICANN assigns the management of these TLDs to these registries
 - These registries, in turn, sell domain names to second level registries
 - The ccTLDs (Country Code TLDs) represent a country of geographic territory.



VERISIGN™



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Second Tier Registries

- Domain resellers
 - Buy domains at a fixed rate from a TLD registry
 - Resell domains based on popularity, bundled into other services, etc.



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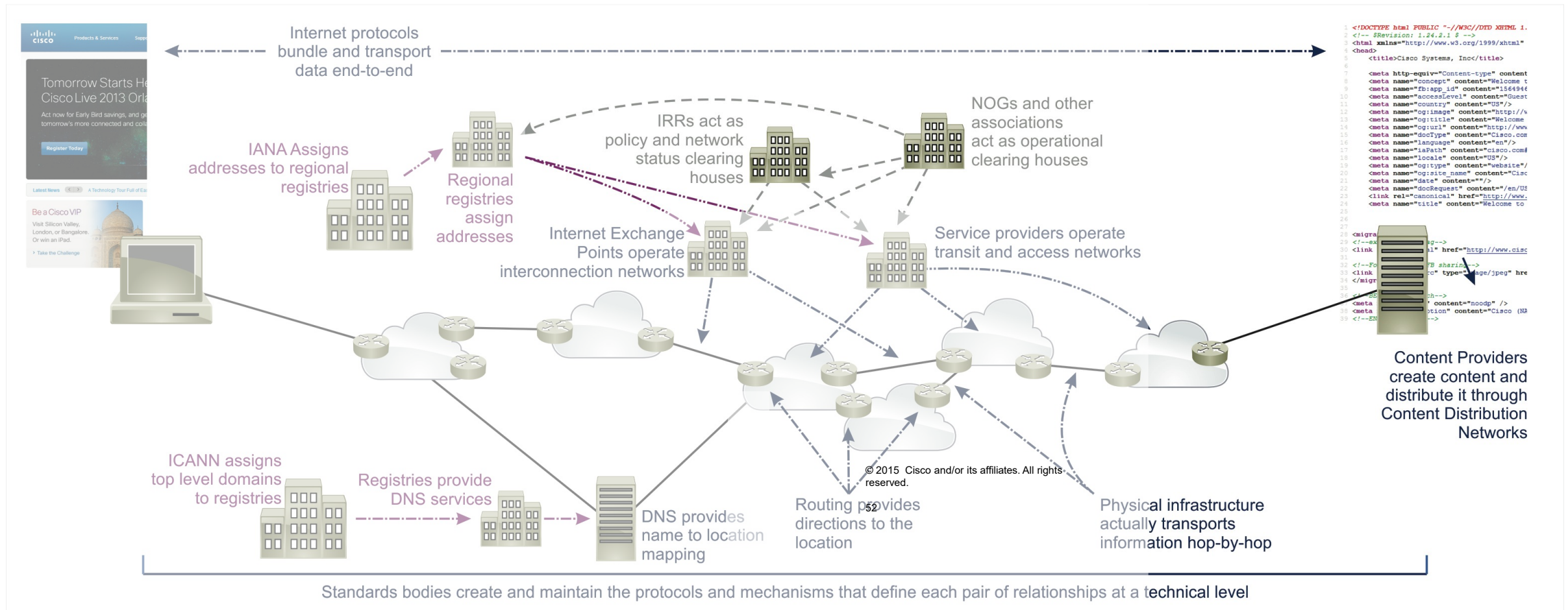
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OTHER ENTITIES

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Clearing Houses



Internet Route Registries

- Internet Routing Registries (IRRs) are cooperative databases
 - Often run through volunteer efforts, by registries, and by research organizations
 - RIPE, APNIC, ARIN, EasyNet, Level3, and others all run IRR mirrors
 - Data is held in a common policy language standardized by the IETF, RPSL
- What is this information used for?
 - A large number of service providers won't accept route advertisements (through BGP) unless there is an existing entry in an IRR they recognize as authoritative
 - Service providers built route filters based on the information in the IRR databases

Internet Route Registries

- The policy information contained in the IRR can include
 - The origin AS
 - The prefix length, longest prefix length, and covered prefixes
 - Filtering policies implemented by a particular AS
 - Whether or not a peering AS is transit or not
- This information is stored in Routing Policy Specification Language (RPSL)
 - RPSL is defined in RFC2622

Network Operators Groups

- Network Operator's Groups (NOGs) act as clearing houses for information about network operations
 - These are generally volunteer organizations
 - Often run in close connection with regional registries, research organizations, vendors, and other regional organizations





The Internet Society engages in a wide spectrum of Internet issues, including policy, governance, technology, and development. We establish and promote principles that are intended to persuade governments to make decisions that are right for their citizens and each nation's future. Everything we do is based on ensuring that a healthy, sustainable Internet is available to everyone – today and for the next billion users.

The Internet Society (<http://www.internetsociety.org>)

Summary

- The Internet is made up of a collection of informal relationships, both related to technology and business.
- It is important to understand the interaction between all the parts and how they complement each other and contribute to a complete system.